LAB # 08

LISTING

OBJECTIVE

Exploring list/arrays in python programming.

THEORY

A list can store a collection of data of any size. It is a collection which is ordered and changeable. The elements in a list are separated by commas and are enclosed by a pair of brackets ***[ ].***

Creating a lists:
A list is a sequence defined by the list class but also have alternative for creation of list without built-in function.

Syntax: With Built-in function list( )

```
list1 = list() # Create an empty list
list2 = list([2, 3, 4]) # Create a list with elements 2, 3, 4
list3 = list(["red", "green", "blue"])#Create a list with strings
list4 = list(range(3, 6)) # Create a list with elements 3, 4, 5
list5 = list("abcd") # Create a list with characters a, b, c, d
```

Without Built-in Function

```
list1 = []                  #Empty list
list2 = [2, 3, 4]           #Integer type list
list3 = ["red", "green"]    #String type list
list4 = [2, "three", 4]     #Mixed data type list
```

Example:

```
#Create list
list_1 = ["apple", "banana", "cherry"]
#display list
print("Current List:",list_1)
```

Accessing items from the list:
An element in a list can be accessed through the ***index operator***, using the following syntax: **myList[index].** List indexes are starts from 0 to len(myList)-1.

Negative indexing means beginning from the end, -1 refers to the last item, -2 refers to the second last item etc.

Example:

```
myList = ["The", "earth", "revolves", "around", "sun"]
print("Postive index:",myList[4])
print("Negative index:",myList[-2])
```

Output:

```
>>> %Run task1.py
Postive index: sun
Negative index: around
```

List Slicing [start : end]:

The index operator allows to select an element at the specified index. The slicing operator returns a slice of the list using the syntax ***list[start : end].*** The slice is a sublist from index start to index end – 1.

Example:

```
list1= [1,2,4,5,6,7,8]
print("Slicing 2 to 5 index:",list1[2:5])
print("Slicing before 3rd index value:",list1[:3])
print("Slicing after 3rd index value:",list1[3:])
```

Output:

```
>>> %Run task2.py
Slicing 2 to 5 index: [4, 5, 6]
Slicing before 3rd index value: [1, 2, 4]
Slicing after 3rd index value: [5, 6, 7, 8]
```

Modifying Elements in a List:

The syntax for modifying an element is similar to the syntax for accessing an element in a list.

Example:

```
#Create a list
motorcycles = ['honda', 'yamaha', 'suzuki']
print(motorcycles)
#Modify list through indexes
motorcycles[0] = 'Swift'
print(motorcycles)
```

Output:

```
>>> %Run task3.py
['honda', 'yamaha', 'suzuki']
['Swift', 'yamaha', 'suzuki']
```

List Methods for Adding , Changing and Removing items:
Python has a set of built-in methods that you can use on lists/arrays.

Method	Description
append( )	Adds an element at the end of the list
copy( )	Returns a copy of the list
count( )	Returns the number of elements with the specified value
index( )	Returns the index of the first element with the specified value
insert( )	Adds an element at the specified position
pop( )	Removes the element at the specified position
remove( )	Removes the first item with the specified value
reverse( )	Reverses the order of the list
sort( )	Sorts the list

EXERCISE

A. Point out the errors, if any, and paste the output also in the following Python programs.

1. Code

```
Def max_list( list ):
    max = list[ 0 ]
    for a is in list:
        elif a > max:
            max = a
    return max
print(max_list[1, 2, -8, 0])
```

Output

2. Code

```
motorcycles = {'honda', 'yamaha', 'suzuki'}
print(motorcycles)
del motorcycles(0)
print(motorcycles)
```

Output:

3. Code

```
Def dupe_v1(x):
  y = []
  for i in x:
    if i not in y:
      y(append(i))
      return y

a = [1,2,3,4,3,2,1]
print a
print dupe_v1(a)
```

Output:

B. What will be the output of the following programs:

1. Code

```
list1= [1,2,4,5,6,7,8]
print("Negative Slicing:",list1[-4:-1])
x = [1, 2, 3, 4, 5, 6, 7, 8, 9]
print("Odd number:", x[::2])
```

Output

2. Code

```
def multiply_list(elements):
    t = 1
    for x in elements:
        t*= x
    return t
print(multiply_list([1,2,9]))
```

Output

3. Code

```
def add(x,lst=[] ):
    if x not in lst:
        lst.append(x)
        return lst

def main():
    list1 = add(2)
    print(list1)
    list2 = add(3, [11, 12, 13, 14])
    print(list2)
main()
```

Output

C. Write Python programs for the following:

1. Write a program that store the names of a few of your friends in a list called ‘names’. Print each person’s name by accessing each element in the list, one at a time.

2. Write a program that make a list that includes at least four people you’d like to invite to dinner. Then use your list to print a message to each person, inviting them to dinner. But one of your guest can’t make the dinner, so you need to send out a new set of invitations. Delete that person on your list, use ***del statement*** and add one more person at the same specified index, use the ***insert( )*** method. Resend the invitation.

3. Write a program that take list = [30, 1, 2, 1, 0], what is the list after applying each of the following statements? Assume that each line of code is independent.

- list.append(40)
- list.remove(1)
- list.pop(1)
- list.pop()
- list.sort()
- list.reverse()

4. Write a program to define a function called ‘printsquare’ with no parameter, take first 7 integer values and compute their square and stored all square values in the list.